

Shubham Sinha

+91 8905169414 | sinhashubham1911@gmail.com | [shubhamsinha1010.github.io](https://github.com/shubhamsinha1010)

[LinkedIn](#) | [GitHub](#) | [LeetCode](#)

Summary

Python backend & AI engineer with 5+ years building search infrastructure, CDC pipelines, and real-time data platforms. Optimized billion-row BI systems on AWS and shipped production NLP and RAG — from catalog classification to grounded retrieval.

Work Experience

Numerator

Jan 2024 – Present

Engineer 1

Remote, India

- Optimized BI query paths on a **billion-row** MySQL store with data associates, cutting heavy analytical reads from minutes to seconds and raising throughput by **50%**.
- Engineered a layered **Django REST Framework** service over **Elasticsearch**, splitting business APIs, search, and ingestion; reduced median API latency by **150ms**.
- Implemented MySQL-to-Elasticsearch **CDC** with pub-sub and **Celery**, replacing overnight reindexes with sync under **60** seconds.
- Automated EC2, RDS, S3, and Lambda provisioning with **Terraform**, cutting environment setup from hours to minutes.
- Fine-tuned **Hugging Face** models with **spaCy** preprocessing for US, CA, and DE catalogs, improving brand/category attribution by **23%**.

Inexture Solutions LLP

Dec 2020 – Dec 2023

Python Developer

Ahmedabad, Gujarat, India

- Delivered **UPI** deposit workflows for a crypto platform serving over **19M** registered users, enabling in-app wallet funding without card rails.
- Launched **Grafana/ELK** payment dashboards with on-call teams, cutting **MTTR** by **40%** through earlier anomaly detection.
- Engineered **RabbitMQ** producers and consumers that decoupled synchronous paths across microservices on a national employment platform.
- Streamlined multi-stage builds and deployments with **Jenkins CI/CD**, removing most manual release steps.

Skills

Languages: Python, SQL, Go, TypeScript

Backend & Data: Django, Django REST Framework, FastAPI, MySQL, PostgreSQL, Elasticsearch, Celery, RabbitMQ, Kafka, Debezium, WebSockets, SQLAlchemy

Cloud & Delivery: AWS, Terraform, Docker, Jenkins, GitHub Actions, Grafana, Prometheus, ELK

AI / NLP: Hugging Face Transformers, spaCy, RAG

Projects

PulseSearch | *Python, FastAPI, Debezium, Kafka, Elasticsearch, RAG*

- Built a real-time AI-native search stack: MySQL **CDC** via **Debezium** + **Kafka** into **Elasticsearch**, with hybrid BM25 + kNN fused by Reciprocal Rank Fusion.
- Shipped grounded **RAG** on a local LLM with citations and freshness guarantees; effectively-once indexing via version-guarded upserts, replayable backfill, and DLQ isolation.

CricLot | *Python, FastAPI, Next.js, WebSockets, SQLAlchemy*

- Built a server-authoritative live cricket auction engine with a lot state machine; host, team, and display consoles stay in sync over a **WebSocket** hub.
- Validated bids against purse, increment, and squad cap before persist/broadcast so concurrent bids cannot corrupt auction state.

Education

Dharmsinh Desai University

Bachelor of Technology in Information Technology (IT)

Nadiad, Gujarat, India

Aug 2017 – May 2021